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Fully distributed time-varying formation tracking control for multiple quadrotor vehicles via finite-time convergent extended state observer

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Abstract This paper investigates a time-varying anti-disturbance formation problem for a group of quadrotor aircrafts with time-varying uncertainties and a directed interaction topology. A novel Finite-Time Convergent Extended State Observer (FTCESO) based fully-distributed formation control scheme is proposed to enhance the disturbance rejection and the formation tracking performances for networked quadrotors. By adopting the hierarchical control strategy, the multi-quadrotor system is separated into two subsystems: the outer-loop cooperative subsystem and the inner-loop attitude subsystem. In the outer-loop subsystem, with the estimation of disturbing forces and uncertain dynamics from FTCESOs, an adaptive consensus theory based cooperative controller is exploited to ensure the multiple quadrotors form and maintain a time-varying pattern relying only on the positions of the neighboring aircrafts. In the inner-loop subsystem, the desired attitude generated by the cooperative control law is stably tracked under a FTCESO-based attitude controller in a finite time. Based on a detailed algorithm to specify the cooperative control protocol, the feasibility condition to achieve the time-varying anti-disturbance formation tracking is derived and the rigorous analysis of the whole closed-loop multi-quadrotor system is given. Some numerical examples are conducted to intuitively demonstrate the effectiveness and the improvements of the proposed control framework.

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1. Introduction

The past few decades witnessed rapid developments of the formation control for Unmanned Aerial Vehicles (UAVs) due to its potential applications in plenty of fields.^{1–3} Different from fixed-wing UAVs, multi-rotor vehicles have many obvious

superiorities such as simple structure, vertical takeoff and landing, hovering ability, and fast maneuverability.⁴ As a representative application of multi-rotor UAVs, the formation issues of multi-quadrotor networks have attracted wide interest from both research and engineering communities.^{5–8}

There are several significant challenges to the multi-quadrotor formation control in practical scenarios. First, quadrotor vehicles are types of in nature underactuated, strong nonlinear coupling and Multi-Input Multi-Output (MIMO)

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systems, which highly complicates the controller design especially in formation control problems.⁶ Besides, throughout the whole formation flight, each aircraft of the multi-quadrotor system is inevitably subject to time-varying disturbances and other adverse factors, and this heavily threatens the flight stability. Last but not least, the realization of the desired formation generally cannot rely on the centralized coordination. This is because it is practically inapplicable for each aircraft to acquire global information in real time especially in the presence of limited sensing capability of sensors and unreliable information interaction. In such a circumstance, the analysis of the formation control problems for multiple quadrotor UAVs becomes much more complicated.

In order to solve the formation control problems for multiple UAVs, an abundance of approaches have been proposed from different perspectives, including leader-follower approach, virtual structure strategy, behavior-based method, finite-time control framework, etc.⁹ In Ref.¹⁰ and Ref.¹¹, the leader-follower based and virtual structure based time-varying formation protocols are proposed respectively for multi-quadrotor systems with undirected switching topologies. The necessary and sufficient feasibility conditions for the realization of formation control are also extracted. However, the rotational subsystem of the quadrotor aircraft is not involved. In Ref.¹² based on the relative weight of desired behavior of each UAV, a behavioral approach is proposed to carry out a waypoint-passing formation flight. The collision avoidance formation problems for multiple UAVs are considered in Ref.¹³ under a unified distributed optimal control scheme. In this framework, an avoidance cost is introduced via an inverse optimal control method, and the formation control and the obstacle avoidance are integrated together. In Ref.¹⁴, based on a new pairwise collision avoidance function of relative positions and velocities, the distributed coordination control of multiple quadrotor aircrafts is studied in three-dimensional space. With a fault-tolerant idea, in Ref.¹⁵, the Fault-Tolerant Formation Control (FTFC) methodology has been conducted to counteract the actuator faults during the formation flight, in which the finite-time convergence of FTFC gains are also considered. In Ref.¹⁶, a virtual target guidance based model predictive control scheme is synthesized for multiple UAVs to achieve the desired formation relying only on the neighboring information. An event-triggered mechanism is also introduced into the formation protocol to reduce the calculative burden. In Ref.¹⁷, the distributed formation control for multi-quadrotor systems is carried out under a nonsmooth consensus algorithm, in which the translational subsystem is tackled by the linear quadratic regulator optimal approach and a global-bounded finite-time attitude controller is proposed for the rotational subsystem. Fully distributed finite-time formation control problems for multiple quadrotor aircrafts are investigated in Ref.¹⁸ based on the homogeneous system theory and finite-time Lyapunov theory. The proposed protocol is comprised of a leader observer, a trajectory controller and an attitude controller which ensure the quadrotors achieve the desired formation in finite-time. However, the aforementioned works rarely take into consideration the internal uncertainties and external disturbances which inevitably exist in practical formation tasks.

In recent years, some researchers have noticed this issue and have some results on anti-disturbance formation control for multi-UAVs published.^{19–21} In Ref.¹⁹, an priori-bounded

adaptive control scheme is proposed to solve the formation control problems of multiple quadrotor UAVs with parametric uncertainties and external disturbances. However, the disturbances considered in Ref.¹⁹ are actually time-invariant because of the third assumption assumed. The three-dimensional anti-disturbance formation tracking of multiple quadrotors is conducted in Ref.²⁰ using a robust consensus-based approach and filtering design technique. Note that the communication graph adopted in Ref.²⁰ is undirected which means that the information interaction among aircrafts is bidirectional and the resource consumption in practice will be unnecessarily doubled. In Ref.²¹, a finite-time neural network disturbance observer based formation control scheme is proposed, in which a distributed finite-time trajectory controller and a finite-time attitude controller are designed using the nonsingular fast terminal sliding mode approach. Under this type of formation control law, the multi-UAV system can achieve time-invariant formation tracking in a finite time. Actually, up to now, the research results on anti-disturbance formation control for multi-quadrotor systems are not very affluent and complete. Thus, it is still necessary and open to further improve the anti-disturbance formation control performance for the multi-quadrotor system in practical applications.

Inspired by the analysis above and partly motivated by the previous works,^{22–26} in this paper, a novel Finite-Time Convergent Extended State Observer (FTCESO) based fully-distributed formation control framework is proposed, by which the time-varying anti-disturbance formation tracking problems for multi-quadrotor systems with directed interaction graph are conducted in a 3-D space. As we know, the Extended State Observer (ESO) is the core part of Active Disturbance Rejection Control (ADRC) scheme which has a strong capability to deal with the uncertain nonlinear systems.^{27–30} While the linear ESO with high-gain characteristic has been widely applied in varieties of applications, it has an undesirable “peaking phenomenon” and can only asymptotically converge in an infinite time interval.³¹ By contrast, the FTCESO not only can alleviate the peaking problem effectively, but also can achieve the convergence in a finite time. Actually, the FTCESO has a better observation performance under the same conditions, which can be verified in the latter simulation section. Although some works have investigated the applications of FTCESOs on single one object,^{26,32} the extension to multi-quadrotor systems is not trivial, especially in the case where the multiple quadrotors need to perform cooperative missions under a partial directed topology.

The proposed formation control framework is designed based on a general backstepping method, and there are two blocks involved in the scheme: a fully-distributed FTCESO-based cooperative controller for the outer-loop subsystem and a finite-time anti-disturbance attitude tracking controller for the inner-loop subsystem. In the first subsystem, with the fully distributed perspective, an adaptive consensus theory^{22,23} is employed to exploit the formation control protocol relying only on the aircraft’s neighboring output information. By adopting distributed FTCESOs, the velocities of the aircrafts, the external time-varying disturbing forces, uncertain nonlinear dynamics and unknown control inputs of the leader are estimated and compensated in the control law in real time. According to the hierarchical control strategy, the desired attitude for the rotational subsystem is generated by the formation control law. For the inner-loop, based on the homogeneous

system theory, finite-time control technique and the FTCESO, an error feedback controller is developed such that each follower's attitude converges to the desired attitude in a finite time in the presence of time-varying disturbing moments and internal nonlinearities. Compared with the previous works, the main contributions are summarized as follows: (A) A novel FTCESO-based fully-distributed formation control scheme is first proposed for the multi-quadrotor systems to enhance the disturbance rejection and cooperative flight performances, and the rigorous theoretical analysis of the whole closed-loop system is given, (B) compared with Refs.^{10,11,33}, the formation protocol designed in this paper does not need the eigenvalue information of Laplacian matrix or any other global information. Moreover, it is actually a type of output feedback control law only requiring the quadrotors' positions to be measurable, which means that the method proposed here is of more practical significance, iii) in comparison to the ESO-based control scheme in Ref.³³, the developed FTCESO-based formation protocol has got remarkable improvements in the disturbance estimation and formation tracking which are clearly demonstrated in the simulations.

The rest of paper is organized as follows. Some notations and problem statement are presented in Section 2. In Section 3, the FTCESO-based fully-distributed formation control framework is thoroughly addressed with theoretical analysis in both outer-loop and inner-loop subsystems. Section 4 demonstrates the effectiveness and improvements by two numerical examples, and the whole work is concluded in Section 5.

2. Preliminaries and problem statement

2.1. Mathematical preliminaries

Some notations are defined in this section. Let $\mathbf{R}$ denote the Euclidean space, $\mathbf{R}^n$ and $\mathbf{R}^{n \times m}$ be a collection of n -dimensional column vectors and a set of $n \times m$ real matrices, respectively. We use $\mathbf{1}$ to denote column vector of ones with scalar 1 as special cases and $\mathbf{0}_{n \times m}$ to denote the $n \times m$ matrix with all entries being 0. Let $\otimes$ denote the Kronecker product and $\|\cdot\|$ denote the Euclidean norm of a vector or the induced matrix 2-norm. I_N represents an identity matrix with dimension N . For a square matrix A , A^T represents its transpose, $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ denote the minimum and the maximum eigenvalue of matrix A , respectively. $\text{diag}\{a_1, a_2, \dots, a_n\}$ represents a diagonal matrix with $a_1, a_2, \dots, a_n$ on the diagonal. A matrix $A = [a_{ij}] \in \mathbf{R}^{n \times n}$ is referred to as a nonsingular M -matrix if all eigenvalues of A have positive real parts and $a_{ij} \leq 0, \forall i \neq j$. The inequality $x > 0$ means that each element of the vector x is positive. The matrix inequality $X > Y$ means that $X - Y$ is positive definite.

A directed graph $G = \{V, \Omega, W\}$ is defined to describe the interaction topology of the multi-quadrotor system. The graph G consists of a set of nodes $V = \{v_1, v_2, \dots, v_N\}$, a set of edges $\Omega \subseteq \{(v_i, v_j) : v_i, v_j \in V\}$, and an adjacency matrix $W = [w_{ij}] \in \mathbf{R}^{N \times N}$ with nonnegative elements w_{ij} . The edge v_{ij} in G is denoted by the ordered pair of nodes (v_i, v_j) . The entries in W are defined that $w_{ji} > 0$ for $v_{ij} \in \Omega$, and $w_{ij} = 0$ otherwise. In addition, $w_{ii} = 0$ for all $i = 1, 2, \dots, N$. The neighbors set of node v_i is described as $\mathcal{Q}_i = \{v_j \in V : v_{ji} \in \Omega\}$. Let $\deg_{in}(v_i) = \sum_{j=1}^N w_{ji}$ be the in-degree of node v_i and

$D = \text{diag}\{\deg_{in}(v_i), i = 1, 2, \dots, N\}$ be the degree matrix of G . Then define the Laplacian matrix $L \in \mathbf{R}^{N \times N}$ of G as $L = D - W$. A directed graph is deemed to have a spanning tree if there is a node called the root such that there is a directed path from this node to every the other node.

2.2. Related definitions and lemmas

Throughout this paper, the following definitions and lemmas will be used.

Definition 1.²³ An agent is called a leader if its corresponding node in the directed graph does not have the incoming edge. An agent is called a follower if its corresponding node in the directed graph has at least one incoming edge.

Lemma 1.³⁴ If p and q are nonnegative real numbers and m and n are positive real numbers such that $(1/m) + (1/n) = 1$, then $pq \leq (p^m/m) + (q^n/n)$.

Definition 2.²⁵ Consider a nonlinear system of the following form

$$\dot{x} = f(x), f(0) = 0, x \in \mathbf{R}^n \quad (1)$$

where $f: D_0 \rightarrow \mathbf{R}^n$ is continuous in an open neighborhood D_0 of the origin. The vector field $f(x) = [f_1(x), f_2(x), \dots, f_n(x)]^T$ is homogenous of degree ϖ with respect to $(\rho_1, \rho_2, \dots, \rho_n)$, if for any given $\delta > 0$,

$$f_i(\delta^{\rho_1} x_1, \delta^{\rho_2} x_2, \dots, \delta^{\rho_n} x_n) = \delta^{\varpi + \rho_i} f_i(x), i \in \{1, 2, \dots, n\}, x \in \mathbf{R}^n$$

where $\varpi > -\min \rho_i$ and $\rho_i > 0, i \in \{1, 2, \dots, n\}$. Then, if $f(x)$ is homogenous, system (1) is said to be homogenous.

Lemma 2.²⁵ If system (1) is homogenous of degree $\varpi < 0$ with respect to $(\rho_1, \rho_2, \dots, \rho_n)$ and $f(x)$ with asymptotically stable equilibrium $x = 0$, the system (1) is globally finite-time stable.

Lemma 3.³⁵ Suppose there exists an open neighborhood D_0 of the origin and a continuous positive definite function $V_v(x) : D_0 \rightarrow \mathbf{R}$ for system (1) such that

$$\dot{V}_v(x) + \vartheta(V_v(x))^\omega \leq 0, x \in o\{0\} \quad (2)$$

where $\vartheta > 0, \omega \in (0, 1)$, and $o \subset D_0$. Then the origin is the finite-time stable equilibrium of system (1). The continuous settling-time function T satisfies

$$T(x) \leq \frac{1}{\vartheta(1-\omega)} (V_v(x))^{1-\omega} \quad (3)$$

Moreover, if D_0 can be chosen to be $\mathbf{R}$, then the origin is said to be globally finite stable equilibrium.

Lemma 4.²⁶ Consider the following second-order uncertain nonlinear system

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = u + f_d(t, x_1, x_2) \\ y = x_1 \end{cases} \quad (4)$$

where $\mathbf{x} = [x_1, x_2]^T \in \mathbf{R}^2$ are the states of system (4), $u \in \mathbf{R}$ is the control input and $f_d(t, x_1, x_2) : \mathbf{R}^3 \rightarrow \mathbf{R}$ is the augmented uncertainty including the uncertain nonlinearities and external disturbances. If $f_d(t, x_1, x_2)$ is uniformly bounded and its time derivative is also bounded by $|\dot{f}_d(t, x_1, x_2)| \leq U_f$, one can denote $x_3 = f_d(t, x_1, x_2)$ as an extended state, and the FTESO can be conducted as

$$\begin{cases} \dot{\hat{x}}_1 = \hat{x}_2 + k_1 \text{sig}^{(\kappa+1)/2}(x_1 - \hat{x}_1) \\ \dot{\hat{x}}_2 = \hat{x}_3 + k_2 \text{sig}^{(\kappa+1)/2}(x_1 - \hat{x}_1) + u \\ \dot{\hat{x}}_3 = k_3 \text{sig}^\kappa(x_1 - \hat{x}_1) \end{cases} \quad (5)$$

where $[\hat{x}_1, \hat{x}_2, \hat{x}_3]^T \in \mathbf{R}^3$ are the estimated states, $\kappa \in (0, 1)$, $k_1, k_2, k_3 > 0$, $\text{sig}^\kappa(\cdot) = \text{sgn}(\cdot)|\cdot|^\kappa$, where $\text{sgn}(\cdot)$ is the standard signum function. Define $\eta_i = x_i - \hat{x}_i$, $i \in \{1, 2, 3\}$, the estimation error system can be obtained

$$\begin{cases} \dot{\eta}_1 = \eta_2 - k_1 \text{sig}^{(\kappa+1)/2}(\eta_1) \\ \dot{\eta}_2 = \eta_3 - k_2 \text{sig}^{(\kappa+1)/2}(\eta_1) \\ \dot{\eta}_3 = x_3 - k_3 \text{sig}^\kappa(\eta_1) \end{cases} \quad (6)$$

For the FTESO above, there exists $t_f > 0$ such that

$$\|\boldsymbol{\eta}\| \leq \frac{\varphi U_f}{\sigma_{\min}\{\mathbf{Q}\}\sigma_{\min}\{\mathbf{E}\}} = \varepsilon_f, \quad \forall t > t_f \quad (7)$$

$$t_f \leq \frac{2(a+b)}{ab} (\lambda_{\max}\{\mathbf{E}\} V_f(\boldsymbol{\eta}_0))^{1/2} \quad (8)$$

where $\sigma_{\min}\{\cdot\}$ represents the minimum singularity of $\{\cdot\}$, estimation error $\boldsymbol{\eta} = [\text{sig}^{(\kappa+1)/2}(\eta_1), \eta_2, \eta_3]^T$, $\boldsymbol{\eta}_0$ denotes the initial estimation error, $V_f(\boldsymbol{\eta}) = \boldsymbol{\eta}^T \mathbf{E} \boldsymbol{\eta}$, $a = (\kappa + 1)\sigma_{\min}\{\mathbf{Q}\}\sigma_{\min}\{\mathbf{E}\} - 2\varphi U_f$, $b = 2\varepsilon_f \sigma_{\min}\{\mathbf{Q}\}\sigma_{\min}\{\mathbf{E}\} - 2\varphi U_f$,

$$\varphi = \sqrt{k_3^2 + 4}, \quad \mathbf{Q} = \begin{bmatrix} k_1 & -1 & 0 \\ k_2 & 0 & -1 \\ k_3 & 0 & 0 \end{bmatrix}, \quad \text{and}$$

$$\mathbf{E} = \begin{bmatrix} 2k_1/(\kappa + 1) + k_2^2 + k_3^2 & -k_2 & -k_3 \\ -k_2 & 2 & 0 \\ -k_3 & 0 & 2 \end{bmatrix}.$$

Note that the parameters k_1, k_2 and k_3 can be selected properly to make $\varphi U_f \ll \sigma_{\min}\{\mathbf{Q}\}\sigma_{\min}\{\mathbf{E}\}$, which means that the estimation error $\boldsymbol{\eta}$ can be arbitrarily sufficiently small in the finite time t_f .

2.3. Quadrotor mathematical model

Consider a multi-quadrotor system consisting of $N - 1$ followers and one leader. Let $F = \{1, 2, \dots, N - 1\}$ be the follower subscript set. The configuration of each quadrotor is shown in Fig. 1. In order to describe the dynamics of each quadrotor, an inertial coordinate system $O_E x_E y_E z_E$ and the body-fixed coordinate systems $O_i x_B y_B z_B$, $i \in F$ are established according to Ref.¹⁸.

The transformation matrix of the i th follower from body coordinate system $O_i x_B y_B z_B$ to inertial coordinate system $O_E x_E y_E z_E$ is given by

$$\mathbf{R}_i^{\mathbf{B} \rightarrow \mathbf{E}} = \begin{bmatrix} c\theta_i c\psi_i & s\theta_i s\phi_i c\psi_i - c\phi_i s\psi_i & c\phi_i s\theta_i c\psi_i + s\phi_i s\psi_i \\ c\theta_i s\psi_i & s\theta_i s\phi_i s\psi_i + c\phi_i c\psi_i & c\phi_i s\theta_i s\psi_i - s\phi_i c\psi_i \\ -s\theta_i & s\phi_i c\theta_i & c\phi_i c\theta_i \end{bmatrix}, \quad i \in F \quad (9)$$

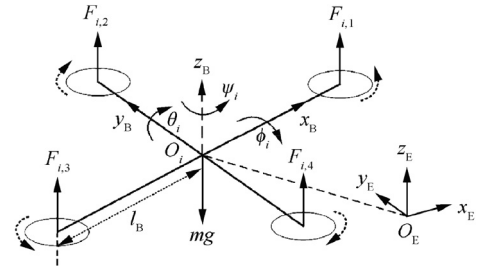


Fig. 1 Configuration of each quadrotor.

where s and c denote $\sin(\cdot)$ and $\cos(\cdot)$. θ_i, ϕ_i, ψ_i represent the pitch angle, roll angle and yaw angle of the i th quadrotor, respectively. It is notable that all the Euler angles are bounded by $(-\pi/2, \pi/2)$ in this paper and the singularity problem is therefor avoided.

The translational movement of the i th follower quadrotor is given by

$$\begin{cases} \dot{\mathbf{p}}_i = \mathbf{v}_i, \quad i \in F \\ \dot{v}_{i,x} = -\frac{C_{Dx}}{m} v_{i,x} + \frac{A_{i,x}}{m} + u_{i,x} \\ \dot{v}_{i,y} = -\frac{C_{Dy}}{m} v_{i,y} + \frac{A_{i,y}}{m} + u_{i,y} \\ \dot{v}_{i,z} = -g - \frac{C_{Dz}}{m} v_{i,z} + \frac{A_{i,z}}{m} + u_{i,z} \end{cases} \quad (10)$$

where

$$\begin{cases} u_{i,x} = \frac{T_i}{m} (\cos\phi_i \sin\theta_i \cos\psi_i + \sin\phi_i \sin\psi_i) \\ u_{i,y} = \frac{T_i}{m} (\cos\phi_i \sin\theta_i \sin\psi_i - \sin\phi_i \cos\psi_i) \\ u_{i,z} = \frac{T_i}{m} (\cos\phi_i \cos\theta_i), \quad i \in F \end{cases} \quad (11)$$

and $\mathbf{p}_i = [p_{i,x}, p_{i,y}, p_{i,z}]^T$ and $\mathbf{v}_i = [v_{i,x}, v_{i,y}, v_{i,z}]^T$, $i \in F$ denote the three-dimensional positions and velocities of the i th follower, respectively. m is the mass of each quadrotor, g is the acceleration of gravity, T_i denotes the total thrust of the i th follower, C_{Dx}, C_{Dy}, C_{Dz} represent the coefficients of drag forces, and $A_{i,x}, A_{i,y}, A_{i,z}$ represent the time-varying disturbing forces in 3-D space for the i th quadrotor.

According to many previous results on the quadrotors,^{5,24,36} and considering the fact that in practice, most quadrotors have relatively slow speed and small attitude angles, and the three control channels of roll, pitch and yaw are usually independent, the rotational dynamics of the i th follower can be simplified as

$$\begin{cases} \ddot{\phi}_i = \dot{\theta}_i \dot{\psi}_i \frac{(J_y - J_z)}{J_x} - \frac{C_{D\phi}}{J_x} \dot{\phi}_i + \frac{M_{i,\phi}}{J_x} + \tau_{i,\phi} \\ \ddot{\theta}_i = \dot{\phi}_i \dot{\psi}_i \frac{(J_z - J_x)}{J_y} - \frac{C_{D\theta}}{J_y} \dot{\theta}_i + \frac{M_{i,\theta}}{J_y} + \tau_{i,\theta} \\ \ddot{\psi}_i = \dot{\phi}_i \dot{\theta}_i \frac{(J_x - J_y)}{J_z} - \frac{C_{D\psi}}{J_z} \dot{\psi}_i + \frac{M_{i,\psi}}{J_z} + \tau_{i,\psi} \end{cases}, \quad i \in F \quad (12)$$

where $\dot{\phi}_i, \dot{\theta}_i, \dot{\psi}_i$ represent the angular rate of three Euler angles and $\ddot{\phi}_i, \ddot{\theta}_i, \ddot{\psi}_i$ represent the angular accelerations. J_x, J_y, J_z are inertial moments of the quadrotor, $C_{D\phi}, C_{D\theta}, C_{D\psi}$ denote the coefficients of drag moments in three directions, $M_{i,\phi}, M_{i,\theta}, M_{i,\psi}$ represent the time-varying disturbing moments, and $\tau_{i,\phi}, \tau_{i,\theta}, \tau_{i,\psi}$ represent the control torques in 3-D space.

According to the aircraft configuration in Fig. 1, the transformation relationship between the control torques $\tau_{i,\phi}, \tau_{i,\theta}, \tau_{i,\psi}$, the total thrust T_i and the actual lift forces $F_{i,1}, F_{i,2}, F_{i,3}, F_{i,4}$ of each follower is given below

$$\begin{bmatrix} T_i \\ \tau_{i,\phi} \\ \tau_{i,\theta} \\ \tau_{i,\psi} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & l_B & 0 & -l_B \\ -l_B & 0 & l_B & 0 \\ -k_c & k_c & -k_c & k_c \end{bmatrix} \begin{bmatrix} F_{i,1} \\ F_{i,2} \\ F_{i,3} \\ F_{i,4} \end{bmatrix}, \quad i \in F \quad (13)$$

where the k_c is torque coefficient, l_B is the distance from each motor to the mass center of the quadrotor.

The dynamics of the leader is described by

$$\begin{cases} \dot{\mathbf{p}}_N = \mathbf{v}_N \\ \dot{v}_{N,x} = -\frac{C_{Dx}}{m} v_{N,x} + \frac{A_{N,x}}{m} + u_{N,x} \\ \dot{v}_{N,y} = -\frac{C_{Dy}}{m} v_{N,y} + \frac{A_{N,y}}{m} + u_{N,y} \\ \dot{v}_{N,z} = -g - \frac{C_{Dz}}{m} v_{N,z} + \frac{A_{N,z}}{m} + u_{N,z} \end{cases} \quad (14)$$

where $\mathbf{p}_N = [p_{N,x}, p_{N,y}, p_{N,z}]^T$, $\mathbf{v}_N = [v_{N,x}, v_{N,y}, v_{N,z}]^T$ and $[u_{N,x}, u_{N,y}, u_{N,z}]^T$ are the positions, velocities and unknown control inputs of the leader, respectively. $A_{N,x}$, $A_{N,y}$, $A_{N,z}$ are unknown time-varying disturbing forces and $u_{N,x}$, $u_{N,y}$, $u_{N,z}$ are the unknown control inputs. Without loss of generality, suppose that the unknown control inputs $[u_{N,x}, u_{N,y}, u_{N,z}]^T$ and their time derivatives $[\dot{u}_{N,x}, \dot{u}_{N,y}, \dot{u}_{N,z}]^T$ are bounded, and the states of the leader are also bounded under $[u_{N,x}, u_{N,y}, u_{N,z}]^T$. Besides, note that the leader in this paper is actually the position and velocity tracking target, thus, only translational dynamics of the leader are considered.

2.4. Problem formulation

Throughout this paper, the directed communication topology among the multi-quadrotor system satisfies the following assumption.

Assumption 1. ¹⁰The topology G contains a directed spanning tree with the leader as the root node.

By Assumption 1, the following results are satisfied.

Lemma 5. ³⁷If and only if G has a directed spanning tree, zero is a simple eigenvalue of $\mathbf{L}$ associated with eigenvector $\mathbf{1}$, and all the other nonzero eigenvalues have positive real parts.

One gets from Definition 1 that the Laplacian matrix $\mathbf{L}$ associated with the directed graph G is as follows

$$\mathbf{L} = \begin{bmatrix} \mathbf{L}_1 & \mathbf{L}_2 \\ \mathbf{0}_{1 \times (N-1)} & \mathbf{0} \end{bmatrix} \quad (15)$$

where $\mathbf{L}_1 \in \mathbf{R}^{(N-1) \times (N-1)}$ and $\mathbf{L}_2 \in \mathbf{R}^{(N-1) \times 1}$. From Assumption 1 and Lemma 5, it follows that the real parts of all eigenvalues of $\mathbf{L}_1$ are positive. Then it can be further obtained that $\mathbf{L}_1$ is a diagonally dominant M -matrix and is nonsingular.

Define the piecewise continuously differentiable vector $\mathbf{h}_{ij}(t) = [h_{ij}^p, h_{ij}^v]^T \in \mathbf{R}^2$, $i \in F$, $j \in \{x, y, z\}$ as the time-varying formation vector of the i th follower in j axis direction. $\mathbf{x}_i = [p_{i,x}, v_{i,x}]^T$, $\mathbf{y}_i = [p_{i,y}, v_{i,y}]^T$, $\mathbf{z}_i = [p_{i,z}, v_{i,z}]^T$, $i \in \{1, 2, \dots, N\}$ are denoted as the states of i th quadrotor in the axis of x , y , z direction, respectively.

Definition 3. The multi-quadrotor system formed by Eqs. (10)–(14) is said to achieve time-varying formation tracking if for any given bounded initial states,

$$\begin{cases} \lim_{t \rightarrow \infty} (\mathbf{x}_i - \mathbf{h}_{i,x} - \mathbf{x}_N) = \mathbf{0} \\ \lim_{t \rightarrow \infty} (\mathbf{y}_i - \mathbf{h}_{i,y} - \mathbf{y}_N) = \mathbf{0}, \quad i \in F \\ \lim_{t \rightarrow \infty} (\mathbf{z}_i - \mathbf{h}_{i,z} - \mathbf{z}_N) = \mathbf{0} \end{cases} \quad (16)$$

Assumption 2. The formation vector $\mathbf{h}_{ij}(t)$ and $\dot{\mathbf{h}}_{ij}^v$, $i \in F$, $j \in \{x, y, z\}$ are bounded with respect to t .

Remark 1. Considering the fact that the positions and velocities of quadrotor systems are bounded and continuous, and the control inputs are also bounded due to the inherent limitations of actuators, it is impossible for the multi-quadrotor system to form a formation specified by the divergent formation vectors $\mathbf{h}_{ij}(t)$ and $\dot{\mathbf{h}}_{ij}^v$, $i \in F$, $j \in \{x, y, z\}$. Thus, the boundedness requirement in Assumption 2 is reasonable.

3. Main results

3.1. Fully-distributed anti-disturbance cooperative control

3.1.1. Protocol design

The translational dynamics for the leader and followers in Eqs. (10) and (14) can be rewritten as

$$\begin{cases} \dot{\mathbf{j}}_i(t) = \mathbf{A}\mathbf{j}_i(t) + \mathbf{B}(u_{ij}(t) + \xi_{ij}(t)) \\ \bar{\mathbf{y}}_{ij}(t) = \mathbf{C}\mathbf{j}_i(t), \quad i \in F, j \in \{x, y, z\} \end{cases} \quad (17)$$

$$\begin{cases} \dot{\mathbf{j}}_N(t) = \mathbf{A}\mathbf{j}_N(t) + \mathbf{B}\xi_{N,j}(t) \\ \bar{\mathbf{y}}_{N,j}(t) = \mathbf{C}\mathbf{j}_N(t), \quad j \in \{x, y, z\} \end{cases} \quad (18)$$

where $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\mathbf{B} = [0, 1]^T$, $\mathbf{C} = [1, 0]$, $\bar{\mathbf{y}}_{ij}(t)$, $i \in \{1, 2, \dots, N\}$, $j \in \{x, y, z\}$, is the output of each translational subsystem and

$$\begin{bmatrix} \xi_{i,x}(t) \\ \xi_{i,y}(t) \\ \xi_{i,z}(t) \end{bmatrix} = \frac{1}{m} \begin{bmatrix} -C_{Dx}v_{i,x} + A_{i,x} \\ -C_{Dy}v_{i,y} + A_{i,y} \\ -mg - C_{Dz}v_{i,z} + A_{i,z} \end{bmatrix}, \quad i \in F \quad (19)$$

$$\begin{bmatrix} \xi_{N,x}(t) \\ \xi_{N,y}(t) \\ \xi_{N,z}(t) \end{bmatrix} = \frac{1}{m} \begin{bmatrix} -C_{Dx}v_{N,x} + A_{N,x} \\ -C_{Dy}v_{N,y} + A_{N,y} \\ -mg - C_{Dz}v_{N,z} + A_{N,z} \end{bmatrix} + \begin{bmatrix} u_{N,x} \\ u_{N,y} \\ u_{N,z} \end{bmatrix} \quad (20)$$

Note that in the considered system (17) and (18), only the position of the quadrotor is assumed to be measurable. The velocity and the total uncertainties need to be estimated online which will be presented below. Besides, given that the finite available energy and the limited changing rate of the states, $\xi_{ij}(t)$ and $\dot{\xi}_{ij}(t)$, $i \in \{1, 2, \dots, N\}$, $j \in \{x, y, z\}$, are supposed to be bounded in the multi-quadrotor system. Assume that $|\dot{\xi}_{ij}(t)| \leq U_\xi$, and according to Lemma 4, the FTCESOs for translational subsystem of each quadrotor can be designed as

$$\begin{cases} \hat{p}_{ij} = \hat{v}_{ij} + k_1 \text{sig}^{(\kappa+1)/2}(p_{ij} - \hat{p}_{ij}) \\ \hat{v}_{ij} = \hat{\xi}_{ij} + k_2 \text{sig}^{(\kappa+1)/2}(p_{ij} - \hat{p}_{ij}) + u_{ij}, \quad i \in F, j \in \{x, y, z\} \\ \hat{\xi}_{ij} = k_3 \text{sig}^\kappa(p_{ij} - \hat{p}_{ij}) \end{cases} \quad (21)$$

$$\begin{cases} \hat{p}_{Nj} = \hat{v}_{Nj} + k_1 \text{sig}^{(\kappa+1)/2}(p_{Nj} - \hat{p}_{Nj}) \\ \hat{v}_{Nj} = \hat{\xi}_{Nj} + k_2 \text{sig}^{(\kappa+1)/2}(p_{Nj} - \hat{p}_{Nj}), \quad j \in \{x, y, z\} \\ \hat{\xi}_{Nj} = k_3 \text{sig}^\kappa(p_{Nj} - \hat{p}_{Nj}) \end{cases} \quad (22)$$

and the estimation error system can be formulated as

$$\begin{cases} \dot{\eta}_{ij}^p = \eta_{ij}^v - k_1 \text{sig}^{(\kappa+1)/2}(\eta_{ij}^p) \\ \dot{\eta}_{ij}^v = \eta_{ij}^\xi - k_2 \text{sig}^{(\kappa+1)/2}(\eta_{ij}^v), \quad i \in \{1, 2, \dots, N\}, j \in \{x, y, z\} \\ \dot{\eta}_{ij}^\xi = \xi_{ij} - k_3 \text{sig}^\kappa(\eta_{ij}^p) \end{cases} \quad (23)$$

where $\eta_{ij}^p = p_{ij} - \hat{p}_{ij}$, $\eta_{ij}^v = v_{ij} - \hat{v}_{ij}$, $\eta_{ij}^\xi = \xi_{ij} - \hat{\xi}_{ij}$, $i \in \{1, 2, \dots, N\}$, $j \in \{x, y, z\}$. Define the estimation error $\bar{\eta}_{ij} = [\text{sig}^{(\kappa+1)/2}(\eta_{ij}^p), \eta_{ij}^v, \eta_{ij}^\xi]^T$, from Lemma 4, it can be obtained that there exists $t_\xi > 0$ such that

$$\begin{aligned} \|\bar{\eta}_{ij}\| &\leq \frac{\varphi U_\xi}{\sigma_{\min}\{\mathbf{Q}\}\sigma_{\min}\{\mathbf{E}\}} = e_\xi, \quad \forall t > t_\xi, \quad i \\ &\in \{1, 2, \dots, N\}, \quad j \in \{x, y, z\} \end{aligned} \quad (24)$$

$$t_\xi \leq \frac{2(\bar{a} + \bar{b})}{\bar{a} \bar{b}} \left(\lambda_{\max}\{\mathbf{E}\} V_\xi(\bar{\eta}_0) \right)^{1/2} \quad (25)$$

where $V_\xi(\bar{\eta}_0) = \bar{\eta}_0^T \mathbf{E} \bar{\eta}_0$, $\bar{\eta}_0$ denotes the maximum initial estimation error of $\bar{\eta}_{ij}$, $\bar{a} = (\kappa + 1)\sigma_{\min}\{\mathbf{Q}\}\sigma_{\min}\{\mathbf{E}\} - 2\varphi U_\xi$, $\bar{b} = 2\varepsilon_\xi \sigma_{\min}\{\mathbf{Q}\}\sigma_{\min}\{\mathbf{E}\} - 2\varphi U_\xi$. It follows that e_ξ can be arbitrarily sufficiently small in the finite time t_ξ by proper choice of parameters k_1 , k_2 and k_3 .

To achieve anti-disturbance formation tracking for multi-quadrotor system, the following fully-distributed cooperative control protocol for the outer-loop subsystem is proposed

$$\begin{cases} u_{ij} = c_{ij}(t) \sigma_{ij}(\xi_{ij}^T \mathbf{P}^{-1} \xi_{ij}) \mathbf{K} \xi_{ij}(t) + \dot{h}_{ij}^v + \hat{\xi}_{Nj} - \hat{\xi}_{ij} \\ \dot{c}_{ij}(t) = \xi_{ij}^T(t) \Xi \xi_{ij}(t), \quad i \in F, j \in \{x, y, z\} \end{cases} \quad (26)$$

where

$\xi_{ij} \triangleq \sum_{k=1}^{N-1} w_{ik} \left((\hat{j}_i - \mathbf{h}_{ij}) - (\hat{j}_k - \mathbf{h}_{kj}) \right) + w_{iN} (\hat{j}_i - \mathbf{h}_{ij} - \hat{j}_N)$, $i \in F$, $j \in \{x, y, z\}$, $\sigma_{ij}(\cdot)$ is a monotonic increasing function that satisfies $\sigma_{ij}(s) \geq 1$ with $s > 0$, $c_{ij}(t)$ is the coupling weight related to the i th follower with $c_{ij}(0) \geq 1$, $i \in F$, $j \in \{x, y, z\}$, $\Xi \in \mathbb{R}^{2 \times 2}$ and $\mathbf{K} \in \mathbb{R}^{1 \times 2}$ denote the feedback gain matrices, $\mathbf{P} \in \mathbb{R}^{2 \times 2}$ is a symmetric positive definite matrix, w_{iN} is the weight of directed topology between the i th follower and the leader. The parameters of protocol (26) will be determined in the following algorithm.

Algorithm 1. The cooperative control law (26) can be specified in 4 steps:

Step 1. There exists a symmetric positive definite matrix $\mathbf{P}$ that satisfies the following LMI

$$\mathbf{A}\mathbf{P} + \mathbf{P}\mathbf{A}^T - 2\mathbf{B}\mathbf{B}^T < 0 \quad (27)$$

Step 2. The gain matrix $\mathbf{K}$ can be specified by

$$\mathbf{K} = -\mathbf{B}^T \mathbf{P}^{-1} \quad (28)$$

Step 3. The gain matrix Ξ can be specified by

$$\Xi = \mathbf{P}^{-1} \mathbf{B} \mathbf{B}^T \mathbf{P}^{-1} \quad (29)$$

Step 4. The monotonically increasing functions $\sigma_{ij}(\xi_{ij}^T \mathbf{P}^{-1} \xi_{ij})$, $i \in F$, $j \in \{x, y, z\}$ can be designed as

$$\sigma_{ij}(\xi_{ij}^T \mathbf{P}^{-1} \xi_{ij}) = \left(1 + \xi_{ij}^T \mathbf{P}^{-1} \xi_{ij} \right)^3, \quad i \in F, j \in \{x, y, z\} \quad (30)$$

Algorithm 1 shows the design procedure of the proposed protocol (26). In the following, the detailed analysis of the anti-disturbance formation tracking for the multi-quadrotor system based on the protocol (26) and Algorithm 1 will be conducted.

3.1.2. Stability analysis

Let $\varsigma_j = [\xi_{1j}^T, \xi_{2j}^T, \dots, \xi_{N-1j}^T]^T$, then

$$\varsigma_j = (\mathbf{L}_1 \otimes \mathbf{I}_2) \begin{bmatrix} \hat{j}_1 - \mathbf{h}_{1j} - \hat{j}_N \\ \hat{j}_2 - \mathbf{h}_{2j} - \hat{j}_N \\ \vdots \\ \hat{j}_{N-1} - \mathbf{h}_{N-1j} - \hat{j}_N \end{bmatrix}, \quad j \in \{x, y, z\} \quad (31)$$

According to Definition 3 and the fact that $\mathbf{L}_1$ is nonsingular and Eq. (24) is satisfied, it can be obtained that the formation tracking issue for multi-quadrotor system is solved if and only if

$$\lim_{t \rightarrow \infty} \varsigma_j(t) = \mathbf{0}, \quad j \in \{x, y, z\} \quad (32)$$

Let $\hat{\xi}_j = [\hat{\xi}_{1j}, \hat{\xi}_{2j}, \dots, \hat{\xi}_{N-1j}]^T$, $\xi_j = [\xi_{1j}, \xi_{2j}, \dots, \xi_{N-1j}]^T$ and $\mathbf{h}_j(t) = [\mathbf{h}_{1j}^T, \mathbf{h}_{2j}^T, \dots, \mathbf{h}_{N-1j}^T]^T$, $j \in \{x, y, z\}$. It follows from Eqs. (17), (18) and (26) that the dynamics of ς_j , $j \in \{x, y, z\}$ satisfies

$$\begin{aligned} \dot{\varsigma}_j &= (\mathbf{I}_{N-1} \otimes \mathbf{A} + \mathbf{L}_1 \tilde{\mathbf{C}}_j \tilde{\sigma}_j \otimes \mathbf{B} \mathbf{K}) \varsigma_j + (\mathbf{L}_1 \otimes \mathbf{A}) \mathbf{h}_j + \mathbf{L}_1 \\ &\quad \otimes (\mathbf{B} \mathbf{B}^T - \mathbf{I}_2) \dot{\mathbf{h}}_j + (\mathbf{L}_1 \otimes \mathbf{B}) (\xi_j - \hat{\xi}_j) + (\mathbf{L}_2 \otimes \mathbf{B}) \\ &\quad \times (\xi_{Nj} - \hat{\xi}_{Nj}) \\ &= (\mathbf{I}_{N-1} \otimes \mathbf{A} + \mathbf{L}_1 \tilde{\mathbf{C}}_j \tilde{\sigma}_j \otimes \mathbf{B} \mathbf{K}) \varsigma_j + (\mathbf{L}_1 \\ &\quad \otimes \mathbf{I}_2) ((\mathbf{I}_{N-1} \otimes \mathbf{A}) \mathbf{h}_j + \mathbf{I}_{N-1} \otimes (\mathbf{B} \mathbf{B}^T - \mathbf{I}_2) \dot{\mathbf{h}}_j) \\ &= (\mathbf{I}_{N-1} \otimes \mathbf{A} + \mathbf{L}_1 \tilde{\mathbf{C}}_j \tilde{\sigma}_j \otimes \mathbf{B} \mathbf{K}) \varsigma_j + (\mathbf{L}_1 \otimes \mathbf{I}_2) \Gamma_j \end{aligned} \quad (33)$$

where

$$\begin{aligned} \tilde{\sigma}_j &\triangleq \text{diag}\{\sigma_{1j}(\xi_{1j}^T \mathbf{P}^{-1} \xi_{1j}), \sigma_{2j}(\xi_{2j}^T \mathbf{P}^{-1} \xi_{2j}), \dots, \sigma_{N-1j}(\xi_{N-1j}^T \mathbf{P}^{-1} \xi_{N-1j})\}, \tilde{\mathbf{C}}_j \triangleq \text{diag}\{c_{1j}, c_{2j}, \dots, c_{N-1j}\}, \\ \Gamma_j &= ([\gamma_{1j}, \gamma_{2j}, \dots, \gamma_{N-1j}]^T - [\chi_{1j}, \chi_{2j}, \dots, \chi_{N-1j}]^T)^T, \\ \gamma_{ij} &= [\mathbf{h}_{ij}^v, 0], \chi_{ij} = [\mathbf{h}_{ij}^p, 0], \quad i \in F, j \in \{x, y, z\}. \end{aligned}$$

Before further addressing the main results, another character of the nonsingular M -matrix $\mathbf{L}_1$ is needed.

Lemma 6. ²³ *There exists a positive diagonal matrix $\mathbf{R} \in \mathbb{R}^{(N-1) \times (N-1)}$ such that $\mathbf{R}\mathbf{L}_1 + \mathbf{L}_1^T \mathbf{R}$ is positive definite and $\mathbf{R}\mathbf{L}_1 + \mathbf{L}_1^T \mathbf{R} \geq \bar{\lambda}_0 \mathbf{I}$, where $\bar{\lambda}_0$ donates the smallest eigenvalue of $\mathbf{R}\mathbf{L}_1 + \mathbf{L}_1^T \mathbf{R}$. One such $\mathbf{R}$ is given by $\text{diag}\{r_1, r_2, \dots, r_{N-1}\}$, where $\mathbf{r} = [r_1, r_2, \dots, r_{N-1}]^T = (\mathbf{L}_1^T)^{-1} \mathbf{1}$.*

Based on the Algorithm 1 and Lemmas 1 and 4–6, the feasibility condition for multi-quadrotor system (10)–(14) to achieve anti-disturbance formation tracking is given in the theorem below.

Theorem 1. For the outer-loop subsystem (17)–(18) of the multi-quadrotor network, consider the fully-distributed FTESO-based cooperative control protocol (26) designed by Algorithm 1 and let Assumptions 1-2 hold. Then for any bounded initial conditions, there exists proper choice of the parameters k_1 , k_2 and k_3 for FTESOs (21)–(22) such that the multi-quadrotor system (10)–(14) achieves anti-disturbance cooperative tracking if the following feasibility condition holds

$$\lim_{t \rightarrow \infty} (h_{ij}^v(t) - \hat{h}_{ij}^p(t)) = 0, \quad i \in F, j \in \{x, y, z\} \quad (34)$$

Proof. Consider the Lyapunov function candidate

$$V(\varsigma_j) = \sum_{i=1}^{N-1} c_{ij} r_i \int_0^{\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij}} \sigma_{ij}(s) ds + \frac{\bar{\lambda}_0}{12} \sum_{i=1}^{N-1} c_{ij} \quad (35)$$

where $i \in F$, $j \in \{x, y, z\}$, $\bar{c}_{ij} = c_{ij} - \mu$ and $\mu > 0$ can be determined later. By taking that $\sigma_{ij}(s) \geq 1$ with $s > 0$ and the fact that $c_{ij}(t) \geq 1$ with $t > 0$, it is easily verified that $V(\varsigma_j) > 0$.

Since condition (34) is satisfied, we have

$$\lim_{t \rightarrow \infty} \Gamma_j(t) = 0$$

Further, by noting that Lemma 6 holds, $\varsigma_j(t)$, $j \in \{x, y, z\}$, is continuous with respect to t and there is no zero matrix among $\tilde{\mathbf{C}}_j$, $\tilde{\boldsymbol{\sigma}}_j$, $\mathbf{R}$, $\mathbf{L}_1$, $\mathbf{P}$ and $\mathbf{B}\mathbf{B}^T$, then as $t \rightarrow \infty$, the following inequality holds

$$\begin{aligned} & \varsigma_j^T (\tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \mathbf{R} \mathbf{L}_1 \otimes \mathbf{P}^{-1}) \Gamma_j \\ & \leq \frac{1}{2} \varsigma_j^T \left(\tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \left(\frac{1}{12} \bar{\lambda}_0 \mathbf{I} \right) \tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \otimes \mathbf{P}^{-1} \mathbf{B}\mathbf{B}^T \mathbf{P}^{-1} \right) \varsigma_j \\ & \leq \frac{1}{2} \varsigma_j^T \left(\tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \left(\mathbf{R} \mathbf{L}_1 + \mathbf{L}_1^T \mathbf{R} - \frac{11}{12} \bar{\lambda}_0 \mathbf{I} \right) \tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \otimes \mathbf{P}^{-1} \mathbf{B}\mathbf{B}^T \mathbf{P}^{-1} \right) \varsigma_j, \\ & j \in \{x, y, z\} \end{aligned} \quad (36)$$

The time derivative of $V(\varsigma_j)$, $j \in \{x, y, z\}$ can be obtained that

$$\begin{aligned} \dot{V}(\varsigma_j) &= 2 \sum_{i=1}^{N-1} c_{ij} r_i \varsigma_{ij} \left(\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij} \right) \varsigma_{ij}^T \mathbf{P}^{-1} \dot{\varsigma}_{ij} + \sum_{i=1}^{N-1} \dot{c}_{ij} r_i \\ & \quad \int_0^{\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij}} \sigma_{ij}(s) ds + \frac{\bar{\lambda}_0}{6} \sum_{i=1}^{N-1} (c_{ij} - \mu) \varsigma_{ij}^T \mathbf{P}^{-1} \mathbf{B}\mathbf{B}^T \mathbf{P}^{-1} \varsigma_{ij} \end{aligned} \quad (37)$$

By noting that Eq. (24) is satisfied and substituting Eqs. (33) and (36), we can properly select k_1 , k_2 , k_3 such that as $t \rightarrow \infty$, the following inequality is satisfied

$$\begin{aligned} & 2 \sum_{i=1}^{N-1} c_{ij} r_i \varsigma_{ij} \left(\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij} \right) \varsigma_{ij}^T \mathbf{P}^{-1} \dot{\varsigma}_{ij} = 2 \varsigma_j^T \left(\tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \mathbf{R} \otimes \mathbf{P}^{-1} \right) \dot{\varsigma}_j \\ &= \varsigma_j^T \left(\tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \mathbf{R} \otimes (\mathbf{P}^{-1} \mathbf{A} + \mathbf{A}^T \mathbf{P}^{-1}) - \tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j (\mathbf{R} \mathbf{L}_1 + \mathbf{L}_1^T \mathbf{R}) \tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \otimes \mathbf{P}^{-1} \mathbf{B}\mathbf{B}^T \mathbf{P}^{-1} \right) \varsigma_j \\ &+ 2 \varsigma_j^T \left(\tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \mathbf{R} \mathbf{L}_1 \otimes \mathbf{P}^{-1} \right) \Gamma_j + 2 \varsigma_j^T \left(\tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \mathbf{R} \mathbf{L}_1 \otimes \mathbf{P}^{-1} \mathbf{B} \right) (\xi_j - \tilde{\xi}_j) \\ &+ 2 \varsigma_j^T \left(\tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \mathbf{R} \mathbf{L}_2 \otimes \mathbf{P}^{-1} \mathbf{B} \right) (\xi_{Nj} - \tilde{\xi}_{Nj}) \\ &\leq \varsigma_j^T \left(\tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \mathbf{R} \otimes (\mathbf{P}^{-1} \mathbf{A} + \mathbf{A}^T \mathbf{P}^{-1}) - \frac{11}{12} \bar{\lambda}_0 \tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j^2 \otimes \mathbf{P}^{-1} \mathbf{B}\mathbf{B}^T \mathbf{P}^{-1} \right) \varsigma_j \end{aligned} \quad (38)$$

Note that the monotonic increasing function $\sigma_{ij}(\cdot)$ satisfies $\sigma_{ij}(s) \geq 1$, $i \in F$, $j \in \{x, y, z\}$ with $s > 0$ and the Lemma 1 holds, it follows that

$$\begin{aligned} & \sum_{i=1}^{N-1} \dot{c}_{ij} r_i \int_0^{\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij}} \sigma_{ij}(s) ds \leq \sum_{i=1}^{N-1} \dot{c}_{ij} r_i \varsigma_{ij} \left(\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij} \right) \varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij} \\ & \leq \sum_{i=1}^{N-1} \frac{\dot{c}_{ij} r_i^3}{3 \bar{\lambda}_0} + \sum_{i=1}^{N-1} \frac{2}{3} \bar{\lambda}_0 \dot{c}_{ij} \sigma_{ij}^{\frac{3}{2}} \left(\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij} \right) \left(\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij} \right)^{\frac{3}{2}} \\ & \leq \sum_{i=1}^{N-1} \frac{\dot{c}_{ij} r_i^3}{3 \bar{\lambda}_0} + \sum_{i=1}^{N-1} \frac{2}{3} \bar{\lambda}_0 \dot{c}_{ij} \sigma_{ij}^{\frac{3}{2}} \left(\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij} \right) \left(1 + \varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij} \right)^{\frac{3}{2}} \\ & = \sum_{i=1}^{N-1} \left(\frac{r_i^3}{3 \bar{\lambda}_0} + \frac{2}{3} \bar{\lambda}_0 \sigma_{ij}^2 \left(\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij} \right) \right) \varsigma_{ij}^T \mathbf{P}^{-1} \mathbf{B}\mathbf{B}^T \mathbf{P}^{-1} \varsigma_{ij} \end{aligned} \quad (39)$$

Substituting Eqs. (38) and (39) into Eq. (37) yields

$$\begin{aligned} \dot{V}(\varsigma_j) &\leq \varsigma_j^T \left[\tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \mathbf{R} \otimes (\mathbf{P}^{-1} \mathbf{A} + \mathbf{A}^T \mathbf{P}^{-1}) \right] \varsigma_j \\ &- \sum_{i=1}^{N-1} \left[\bar{\lambda}_0 \left(\frac{11}{12} c_{ij}^2 \sigma_{ij}^2 \left(\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij} \right) \right. \right. \\ &\quad \left. \left. - \frac{1}{6} c_{ij} - \frac{2}{3} \sigma_{ij}^2 \left(\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij} \right) \right) + \frac{1}{6} \left(\bar{\lambda}_0 \mu - \frac{2 r_i^3}{\bar{\lambda}_0} \right) \right] \\ &\quad \times \varsigma_{ij}^T \mathbf{P}^{-1} \mathbf{B}\mathbf{B}^T \mathbf{P}^{-1} \varsigma_{ij}, \quad j \in \{x, y, z\} \end{aligned} \quad (40)$$

Let $\mu \geq \frac{1}{2} \bar{\mu} + \max_{i \in F} \left(2 r_i^3 / \bar{\lambda}_0 \right)$, where $\bar{\mu}$ is a positive scalar which is specified later above Eq. (42). Further, taking the properties of $\sigma_{ij}(\cdot)$ and $c_{ij}(t)$, $j \in \{x, y, z\}$, it follows from Eq. (40) that

$$\begin{aligned} \dot{V}(\varsigma_j) &\leq \varsigma_j^T \left(\tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \mathbf{R} \otimes (\mathbf{P}^{-1} \mathbf{A} + \mathbf{A}^T \mathbf{P}^{-1}) \right) \varsigma_j \\ &- \frac{\bar{\lambda}_0}{12} \sum_{i=1}^{N-1} \left(c_{ij}^2 \sigma_{ij}^2 \left(\varsigma_{ij}^T \mathbf{P}^{-1} \varsigma_{ij} \right) + \bar{\mu} \right) \varsigma_{ij}^T \mathbf{P}^{-1} \mathbf{B}\mathbf{B}^T \mathbf{P}^{-1} \varsigma_{ij} \\ &\leq \varsigma_j^T \left(\tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \mathbf{R} \otimes (\mathbf{P}^{-1} \mathbf{A} + \mathbf{A}^T \mathbf{P}^{-1}) - \frac{1}{6} \sqrt{\bar{\mu} \bar{\lambda}_0} \tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \otimes \mathbf{P}^{-1} \mathbf{B}\mathbf{B}^T \mathbf{P}^{-1} \right) \varsigma_j \end{aligned} \quad (41)$$

Let $\tilde{\xi}_j = \left(\sqrt{\bar{\mu} \bar{\lambda}_0} \tilde{\mathbf{C}}_j \tilde{\boldsymbol{\sigma}}_j \mathbf{R} \otimes \mathbf{I} \right) \varsigma_j$, $j \in \{x, y, z\}$ and select $\bar{\mu}$ to be large enough such that $\sqrt{\bar{\mu} \bar{\lambda}_0} \mathbf{R}^{-1} \geq 12 \mathbf{I}$. Then, it follows from Eq. (41) that

$$\dot{V}(\varsigma_j) \leq \tilde{\xi}_j^T (I_{N-1} \otimes (\mathbf{P}^{-1} \mathbf{A} + \mathbf{A}^T \mathbf{P}^{-1} - 2 \mathbf{P}^{-1} \mathbf{B}\mathbf{B}^T \mathbf{P}^{-1})) \tilde{\xi}_j \leq 0 \quad (42)$$

From Eq. (42), it is verified that ς_j asymptotically converges to 0 as $t \rightarrow \infty$ in the case where the parameters k_1 , k_2 , k_3 of FTCESOs are properly selected and the feasibility condition (34) holds. Thus, the anti-disturbance formation tracking for outer-loop subsystem (17)-(18) of the multi-quadrotor network (10)-(14) is achieved under the fully distributed FTCESO-based cooperative control protocol (26). Prove up.

Remark 2. From Theorem 1, it is known that $\dot{V}(\varsigma_j) \leq 0$ for $\forall t \in [0, \infty)$ which means that $V(\varsigma_j)$, $j \in \{x, y, z\}$ is bounded and so is each $c_{ij}(t)$, $i \in F$, $j \in \{x, y, z\}$. By taking that $\dot{c}_{ij}(t) \geq 0$, it can be verified that the coupling weights $c_{ij}(t)$ are certainly convergent to some finite values.

Remark 3. The proposed FTCESO-based formation protocol relies only on the neighboring aircraft's positions without any requirement of global information like the minimum eigenvalue of the Laplacian matrix or Laplacian matrix itself. Besides, the proposed protocol (26) is actually a type of output-feedback control law, which means that the velocities of the quadrotors do not have to be measurable. The velocities and the uncertain disturbing forces can be estimated by the FTCESOs (21)-(22) in a finite time. It is also noteworthy that the feasibility condition (34) actually has the practical meaning with the derivative of position being equal to the velocity.

Remark 4. As mentioned in Theorem 1, the initial conditions of the multi-quadrotor system are required to be bounded. Actually, this is the requirement for the finite-time convergence of the FTCESOs. Considering the fact that the initial positions of the quadrotors are bounded in practical scenarios, this requirement is reasonable. And note that the bounds of the initial states mentioned here can be unknown and arbitrarily large. Therefore, according to the stability theory of nonlinear systems, the results obtained in this paper are theoretically semi-global.

3.2. Finite-time anti-disturbance attitude tracking

In this part, the finite-time anti-disturbance attitude controller for each quadrotor is derived to track the desired attitude generated by the virtual control forces u_{ij} , $i \in F$, $j \in \{x, y, z\}$. According to Eq. (11), the desired attitude is obtained that

$$\begin{cases} T_i^d = m\sqrt{u_{i,x}^2 + u_{i,y}^2 + u_{i,z}^2} \\ \phi_i^d = \arcsin\left(\frac{u_{i,x}\sin\psi_i^d - u_{i,y}\cos\psi_i^d}{\sqrt{u_{i,x}^2 + u_{i,y}^2 + u_{i,z}^2}}\right) \\ \theta_i^d = \arctan\left(\frac{u_{i,x}\cos\psi_i^d + u_{i,y}\sin\psi_i^d}{u_{i,z}}\right) \end{cases} \quad (43)$$

where ψ_i^d can be generated by some heading sensors to track the trajectory direction or can be simply set to 0 for convenience.

From Eq. (12), the attitude dynamics of the i th follower aircraft can be rewritten as

$$\begin{cases} \dot{j}_i(t) = A j_i(t) + B(\tau_{ij}(t) + \zeta_{ij}(t)) \\ \tilde{y}_{ij}(t) = C j_i(t), \quad i \in F, j \in \{\phi, \theta, \psi\} \end{cases} \quad (44)$$

where $\tilde{y}_{ij}(t)$, $i \in F$, $j \in \{\phi, \theta, \psi\}$, is the output of each attitude subsystem and

$$\begin{bmatrix} \zeta_{i,\phi}(t) \\ \zeta_{i,\theta}(t) \\ \zeta_{i,\psi}(t) \end{bmatrix} = \begin{bmatrix} (\dot{\theta}_i \dot{\psi}_i (J_y - J_z) - C_{D\phi} \dot{\phi}_i + M_{i,\phi})/J_x \\ (\dot{\phi}_i \dot{\psi}_i (J_z - J_x) - C_{D\theta} \dot{\theta}_i + M_{i,\theta})/J_y \\ (\dot{\phi}_i \dot{\theta}_i (J_x - J_y) - C_{D\psi} \dot{\psi}_i + M_{i,\psi})/J_z \end{bmatrix}, \quad i \in F \quad (45)$$

It is notable that in the considered inner-loop system (44), only the attitude angle is assumed to be measurable. The angular velocity and the total uncertainties will be estimated online. In addition, without loss of generality, $\zeta_{ij}(t)$ and $\dot{\zeta}_{ij}(t)$ are also assumed to be bounded and $|\dot{\zeta}_{ij}(t)| \leq U_\zeta$, $i \in F$, $j \in \{\phi, \theta, \psi\}$.

By Lemma 4, the FTCESO for the attitude subsystem (44) is designed as

$$\begin{cases} \hat{j}_{i,1} = \hat{j}_{i,2} + k_1 \text{sig}^{(\kappa+1)/2}(\hat{j}_{i,1} - \hat{j}_{i,2}) \\ \hat{j}_{i,2} = \hat{\zeta}_{ij} + k_2 \text{sig}^{(\kappa+1)/2}(\hat{j}_{i,1} - \hat{j}_{i,2}) + \tau_{ij}, \quad i \in F, j \in \{\phi, \theta, \psi\} \\ \hat{\zeta}_{ij} = k_3 \text{sig}^\kappa(\hat{j}_{i,1} - \hat{j}_{i,2}) \end{cases} \quad (46)$$

where $\hat{j}_{i,1} \rightarrow j_i$, $\hat{j}_{i,2} \rightarrow \dot{j}_i$, $i \in F$, $j \in \{\phi, \theta, \psi\}$.

The dynamics of the estimation error for FTCESO (46) are formulated as

$$\begin{cases} \dot{\eta}_{ij}^\theta = \eta_{ij}^\theta - k_1 \text{sig}^{(\kappa+1)/2}(\eta_{ij}^\theta) \\ \dot{\eta}_{ij}^\zeta = \eta_{ij}^\zeta - k_2 \text{sig}^{(\kappa+1)/2}(\eta_{ij}^\zeta), \quad i \in F, j \in \{\phi, \theta, \psi\} \\ \dot{\eta}_{ij}^\tau = \zeta_{ij} - k_3 \text{sig}^\kappa(\eta_{ij}^\tau) \end{cases} \quad (47)$$

where $\eta_{ij}^\theta = j_{i,1} - \hat{j}_{i,1}$, $\eta_{ij}^\zeta = j_{i,2} - \hat{j}_{i,2}$, $\eta_{ij}^\tau = \zeta_{ij} - \hat{\zeta}_{ij}$, $i \in F$, $j \in \{\phi, \theta, \psi\}$. Define the estimation error of FTCESO (46) as $\tilde{\eta}_{ij} = [\text{sig}^{(\kappa+1)/2}(\eta_{ij}^\theta), \eta_{ij}^\zeta, \eta_{ij}^\tau]^T$, from Lemma 4, it also follows that there exists $t_\zeta > 0$ such that

$$\|\tilde{\eta}_{ij}\| \leq \frac{\varphi U_\zeta}{\sigma_{\min}\{Q\}\sigma_{\min}\{E\}} = \varepsilon_\zeta, \quad \forall t > t_\zeta, \quad i \in F, j \in \{\phi, \theta, \psi\} \quad (48)$$

$$t_\zeta \leq \frac{2(\tilde{a} + \tilde{b})}{\tilde{a}\tilde{b}} \left(\lambda_{\max}\{E\} V_\zeta(\tilde{\eta}_0) \right)^{1/2} \quad (49)$$

where $V_\zeta(\tilde{\eta}_0) = \tilde{\eta}_0^T E \tilde{\eta}_0$, $\tilde{\eta}_0$ denotes the maximum initial estimation error of $\tilde{\eta}_{ij}$, $\tilde{a} = (\kappa + 1)\sigma_{\min}\{Q\}\sigma_{\min}\{E\} - 2\varphi U_\zeta$, $\tilde{b} = 2\varepsilon_\zeta \sigma_{\min}\{Q\}\sigma_{\min}\{E\} - 2\varphi U_\zeta$. It follows that ε_ζ can be arbitrarily sufficiently small in the finite time t_ζ by proper choice of parameters k_1 , k_2 and k_3 .

For subsystem (44) tracking the desired attitude j_i^d , $i \in F$, $j \in \{\phi, \theta, \psi\}$ in a finite time, the FTCESO-based finite-time attitude tracking controller is proposed

$$\tau_{ij} = -\alpha_1 \text{sig}^{\beta_1}(\hat{j}_{i,1} - j_i^d) - \alpha_2 \text{sig}^{\beta_2}(\hat{j}_{i,2} - \dot{j}_i^d) - \hat{\zeta}_{ij} + \dot{j}_i^d \quad (50)$$

where $\alpha_1 > 0$, $\alpha_2 > 0$, $\beta_1 \in (0, 1)$, $\beta_2 = 2\beta_1/(1 + \beta_1) \in (0, 1)$ are control parameters.

Theorem 2. Consider the inner-loop subsystem (44) of multi-quadrotor system (10)-(14) with external disturbing moments and uncertain nonlinearities. Then with the proposed controller (50), the desired attitude from Eq. (43) can be tracked within a finite time interval.

Proof. Substitute the control law (50) into attitude system (44) and it follows that

$$\ddot{j}_i = -\alpha_1 \text{sig}^{\beta_1}(\hat{j}_{i,1} - j_i^d) - \alpha_2 \text{sig}^{\beta_2}(\hat{j}_{i,2} - j_i^d) + \ddot{j}_i^d + \zeta_{ij} - \hat{\zeta}_{ij}, \quad (51)$$

$$i \in F, j \in \{\phi, \theta, \psi\}$$

From Eq. (47), (51) can be further rewritten as

$$\ddot{j}_i = -\alpha_1 \text{sig}^{\beta_1}(j_{i,1} - \eta_{ij}^\Theta - j_i^d) - \alpha_2 \text{sig}^{\beta_2}(j_{i,2} - \eta_{ij}^\Omega - j_i^d) + \ddot{j}_i^d + \eta_{ij}^\zeta \quad (52)$$

Define $e_1 = j_{i,1} - j_i^d$, $e_2 = j_{i,2} - j_i^d$, $\dot{e}_2 = \ddot{j}_i - \ddot{j}_i^d$, then we have

$$\begin{cases} \dot{e}_1 = e_2 \\ \dot{e}_2 = -\alpha_1 \text{sig}^{\beta_1}(e_1 - \eta_{ij}^\Theta) - \alpha_2 \text{sig}^{\beta_2}(e_2 - \eta_{ij}^\Omega) + \eta_{ij}^\zeta \end{cases} \quad (53)$$

To prove Theorem 2, we only have to verify that system (53) converges into a sufficiently small neighborhood of the origin in a finite time.

Let $\bar{e}_1 = e_1 - \eta_{ij}^\Theta$, $\bar{e}_2 = e_2 - \eta_{ij}^\Omega$ and take Eq. (47), then the system (53) is transformed to

$$\begin{cases} \dot{\bar{e}}_1 = \bar{e}_2 + \Delta_1 \\ \dot{\bar{e}}_2 = -\alpha_1 \text{sig}^{\beta_1}(\bar{e}_1) - \alpha_2 \text{sig}^{\beta_2}(\bar{e}_2) + \Delta_2 \end{cases} \quad (54)$$

where $\Delta_1 = k_1 \text{sig}^{(\kappa+1)/2}(\eta_{ij}^\Theta)$, $\Delta_2 = k_2 \text{sig}^{(\kappa+1)/2}(\eta_{ij}^\Omega)$. By Eqs. (47)-(49), it follows that Δ_1 and Δ_2 are convergent in the finite time t_ζ with $|\Delta_1| = k_1 |\text{sig}^{(\kappa+1)/2}(\eta_{ij}^\Theta)| \leq k_1 \varepsilon_\zeta$ and $|\Delta_2| = k_2 |\text{sig}^{(\kappa+1)/2}(\eta_{ij}^\Omega)| \leq k_2 \varepsilon_\zeta$. Thus, Δ_1 and Δ_2 can be actually regarded as slight disturbances of the system (54).

To proceed, at first, we consider the nominal system of (54)

$$\begin{cases} \dot{\bar{e}}_1 = \bar{e}_2 \\ \dot{\bar{e}}_2 = -\alpha_1 \text{sig}^{\beta_1}(\bar{e}_1) - \alpha_2 \text{sig}^{\beta_2}(\bar{e}_2) \end{cases} \quad (55)$$

Define the candidate Lyapunov function of system (55) as follows:

$$V(\bar{e}_1, \bar{e}_2) = \frac{\alpha_1}{1+\beta_1} |\bar{e}_1|^{1+\beta_1} + \frac{1}{2} \bar{e}_2^2 \quad (56)$$

The time derivative of $V(\bar{e}_1, \bar{e}_2)$ is obtained that

$$\begin{aligned} \dot{V}(\bar{e}_1, \bar{e}_2) &= \alpha_1 \text{sig}^{\beta_1}(\bar{e}_1) \bar{e}_2 + \bar{e}_2 \bar{e}_2 = -\alpha_2 \bar{e}_2 \text{sig}^{\beta_2}(\bar{e}_2) \\ &= -\alpha_2 |\bar{e}_2|^{\beta_2+1} \leq 0 \end{aligned} \quad (57)$$

where we have used the fact that $d(|\bar{e}_1|^{1+\beta_1})/dt = (1+\beta_1) \text{sig}^{\beta_1}(\bar{e}_1) \bar{e}_2$.

From Eqs. (55) and (57), it is clear that the nominal system (55) is asymptotically stable. Besides, by Definition 2, subsystem (55) is obviously homogeneous of degree

$\varpi = (\beta_1 - 1)/2 < 0$ related to (ρ_1, ρ_2) with $\rho_1 = 1$ and $\rho_2 = (1 + \beta_1)/2$. Thus, according to Lemma 2, the nominal system (55) globally converges to zero in a finite time.

In the following, the settling time t_e is clearly given. From Eqs. (56) and (57), it can be easily verified that, for $\delta > 0$,

$$V(\delta^{1/\beta_1} \bar{e}_1, \delta^{1/\beta_2} \bar{e}_2) = \delta^{(1+\beta_1)/\beta_1} V(\bar{e}_1, \bar{e}_2) \quad (58)$$

and

$$\dot{V}(\delta^{1/\beta_1} \bar{e}_1, \delta^{1/\beta_2} \bar{e}_2) = \delta^{(1+\beta_2)/\beta_2} \dot{V}(\bar{e}_1, \bar{e}_2) \quad (59)$$

Now define

$$\vartheta = -\max_{(e^{-1}, e^{-2}) \in \Pi} \dot{V}(\bar{e}_1, \bar{e}_2) \quad (60)$$

where $\Pi = \{(\bar{e}_1, \bar{e}_2) : V(\bar{e}_1, \bar{e}_2) = 1\}$, note that $\vartheta > 0$. Then, let $\delta = (V(\bar{e}_1, \bar{e}_2))^{-\beta_1/(1+\beta_1)}$, and from Eq. (59), we have

$$\frac{\dot{V}(\bar{e}_1, \bar{e}_2)}{(V(\bar{e}_1, \bar{e}_2))^{(1+3\beta_1)/(2+2\beta_1)}} = \dot{V}((V(\bar{e}_1, \bar{e}_2))^{-1/(1+\beta_1)} \bar{e}_1, (V(\bar{e}_1, \bar{e}_2))^{-1/2} \bar{e}_2) \quad (61)$$

From Eq. (58), it follows that $V((V(\bar{e}_1, \bar{e}_2))^{-1/(1+\beta_1)} \bar{e}_1, (V(\bar{e}_1, \bar{e}_2))^{-1/2} \bar{e}_2) = 1$, which means that

$$\{((V(\bar{e}_1, \bar{e}_2))^{-1/(1+\beta_1)} \bar{e}_1, (V(\bar{e}_1, \bar{e}_2))^{-1/2} \bar{e}_2) : (\bar{e}_1, \bar{e}_2) \in \mathbf{R}^2 \setminus \{0\}\} = \Pi \quad (62)$$

According to Eqs. (61) and (62), it can be obtained that

$$\begin{aligned} &\sup_{(e^{-1}, e^{-2}) \neq (0,0)} \frac{\dot{V}(\bar{e}_1, \bar{e}_2)}{(V(\bar{e}_1, \bar{e}_2))^{(1+3\beta_1)/(2+2\beta_1)}} \\ &= \sup_{(e^{-1}, e^{-2}) \neq (0,0)} \dot{V}((V(\bar{e}_1, \bar{e}_2))^{-1/(1+\beta_1)} \bar{e}_1, (V(\bar{e}_1, \bar{e}_2))^{-1/2} \bar{e}_2) \\ &= \sup_{(e^{-1}, e^{-2}) \in \Pi} \dot{V}(\bar{e}_1, \bar{e}_2) = -\vartheta \end{aligned} \quad (63)$$

Therefore, we have

$$\dot{V}(\bar{e}_1, \bar{e}_2) \leq -\vartheta (V(\bar{e}_1, \bar{e}_2))^{(1+3\beta_1)/(2+2\beta_1)} \quad (64)$$

Since $\beta_1 \in (0, 1)$, $(1 + 3\beta_1)/(2 + 2\beta_1) \in (0, 1)$, and from Lemma 3, the settling time t_e can be obtained as

$$\begin{aligned} t_e &\leq \frac{2(1+\beta_1)}{\vartheta(1-\beta_1)} (V(\bar{e}_1, \bar{e}_2))^{(1-\beta_1)/(1+\beta_1)} \\ &\leq \frac{2(1+\beta_1)}{\vartheta(1-\beta_1)} (V(\bar{e}_{1,0}, \bar{e}_{2,0}))^{(1-\beta_1)/(1+\beta_1)} \end{aligned} \quad (65)$$

where $\bar{e}_{1,0}$ and $\bar{e}_{2,0}$ are the maximum initial values of $\bar{e}_1$ and $\bar{e}_2$, respectively. From the analysis above, the subsystem (55) can asymptotically converge in a finite time t_e .

Next, the finite-time convergence of system (54) will be proved. Define $\gamma = [\bar{e}_1, \bar{e}_2]^T$ and rewrite the system (54) as

$$\dot{\gamma} = \bar{A} \gamma + \bar{B} \quad (66)$$

$$\text{where } \bar{A} = \begin{bmatrix} 0 & 1 \\ -\alpha_1 |\bar{e}_1|^{\beta_1-1} & -\alpha_2 |\bar{e}_2|^{\beta_2-1} \end{bmatrix}, \bar{B} = \begin{bmatrix} \Delta_1 \\ \Delta_2 \end{bmatrix}.$$

Since $\alpha_1, \alpha_2 > 0$, $\bar{\mathbf{A}}$ is Hurwitz except $\gamma = \mathbf{0}$. Therefore, it can be obtained that³⁸

$$\|\gamma\| \leq \frac{\|\bar{\mathbf{B}}\|}{\sigma_{\min}\{\bar{\mathbf{A}}\}} = \frac{\sqrt{\Delta_1^2 + \Delta_2^2}}{\sigma_{\min}\{\bar{\mathbf{A}}\}} \leq \frac{\sqrt{k_1^2 + k_2^2}}{\sigma_{\min}\{\bar{\mathbf{A}}\}} \varepsilon_\zeta \quad (67)$$

where $\sigma_{\min}\{\bar{\mathbf{A}}\}$ represents the minimum singularity of $\bar{\mathbf{A}}$.

Note that, since $|\bar{e}_1|^{\beta_1-1}$ and $|\bar{e}_2|^{\beta_2-1}$ are monotonic decreasing functions of $\bar{e}_1$ and $\bar{e}_2$, $\sigma_{\min}\{\bar{\mathbf{A}}\}$ is also monotonic decreasing function of $\|\gamma\|$. That is, system (66) is at least asymptotically convergent. Meanwhile, considering the finite-time convergence of nominal system (55) and FTCESOs (46)–(47), it can be concluded that the system (66) is also finite-time convergent. That is, the subsystem (53) is globally finite-time stable, and the total settling time of the attitude subsystem (44) is

$$\begin{aligned} t_{\text{total}} &\leq t_\zeta + t_e \\ &\leq \frac{2(\tilde{a} + \tilde{b})}{\tilde{a}\tilde{b}} \left(\lambda_{\max}\{\mathbf{E}\} V_\zeta(\tilde{\eta}_0) \right)^{1/2} \\ &\quad + \frac{2(1 + \beta_1)}{\vartheta(1 - \beta_1)} (V(\bar{e}_{1,0}, \bar{e}_{2,0}))^{(1-\beta_1)/(1+\beta_1)} \end{aligned} \quad (68)$$

Thus, Theorem 2 is proved.

Remark 5. The proposed finite-time FTCESO-based attitude controller can converge the attitude of each quadrotor to the desired attitude in finite-time under the time-varying disturbing moments and uncertain nonlinearities. It is notable that the designed controller is developed on the error feedback control framework with the finite-time control technique, homogeneous system theory and the ADRC methodology. If $\beta_1 = 1$, the control law (50) will straightforward become a feedforward compensating PD controller.

4. Simulation

In this section, the simulations are conducted to demonstrate the effectiveness and innovations of the theoretical results. Consider a multi-quadrotor system of six followers and one leader with the dynamics (10)–(14).

The information exchange among the multiple quadrotors is described by a 0-1 weighted directed topology $\mathbf{G}$ as shown in Fig. 2. It can be verified that the Assumption 1 is satisfied. The model parameters of each quadrotor are given in Table 1.

4.1. Effectiveness verification

To verify the effectiveness of the proposed FTCESO-based fully-distributed formation tracking scheme, six follower quadrotors are supposed to track the leader quadrotor and form a spinning parallel hexagram formation at the same time. The time-varying formation vector is denoted as $\mathbf{h}_i(t) = [h_{i,x}^p, h_{i,x}^v, h_{i,y}^p, h_{i,y}^v, h_{i,z}^p, h_{i,z}^v]^T$, where $i \in \{1, 2, \dots, 6\}$ and $h_{i,x}^p = -5\cos(t + (i-1)\pi/3)$, $h_{i,x}^v = 5\sin(t + (i-1)\pi/3) + e^{-2t}$,

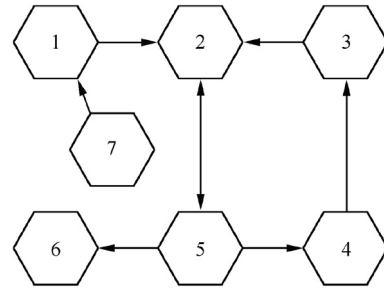


Fig. 2 Directed interaction topology $\mathbf{G}$.

Table 1 Model parameters of each quadrotor.

Parameter	Value
$m(\text{kg})$	1
$g(\text{m} \cdot \text{s}^{-2})$	9.8
$J_x, J_y(\text{kg} \cdot \text{m}^2)$	0.03
$J_z(\text{kg} \cdot \text{m}^2)$	0.06
$C_{Dx}, C_{Dy}(\text{kg} \cdot \text{s}^{-1})$	0.1
$C_{Dz}(\text{kg} \cdot \text{s}^{-1})$	0.08
$C_{D\phi}, C_{D\theta}(\text{kg} \cdot \text{m} \cdot \text{s}^{-1})$	0.03
$C_{D\psi}(\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-1})$	0.05
$k_c(\text{m})$	0.06
$l_B(\text{m})$	0.33

$h_{i,y}^p = 5\sin(t + (i-1)\pi/3)$, $h_{i,y}^v = 5\cos(t + (i-1)\pi/3) + 2e^{-2t}$, $h_{i,z}^p = 3\cos(t + (i-1)\pi/3)$, $h_{i,z}^v = -3\sin(t + (i-1)\pi/3) + 3e^{-2t}$. The external time-varying disturbing forces are described as $A_{i,j} = 0.3(\sin(t) + \cos(t))/i$, $i \in \{1, 2, \dots, 7\}$, $j \in \{x, y, z\}$ and the external time-varying disturbing moments are denoted as $M_{i,j} = 0.1\cos(it)$, $i \in \{1, 2, \dots, 6\}$, $j \in \{\phi, \theta, \psi\}$. The unknown control inputs of the leader are given as $u_{7,x} = \sin(3t) + 0.5$, $u_{7,y} = 2\sin^2(t) - 1.5$, and $u_{7,z} = \cos(2t) + 10.5$.

According to Ref.²⁶, the parameters for FTCESOs (21), (22) and (46) are selected as $k_1 = 30$, $k_2 = 790$, $k_3 = 980$ and $\kappa = 0.35$. By Algorithm 1, the gain matrices for outer-loop cooperative control protocol (26) can be obtained as

$$\mathbf{P} = \begin{bmatrix} 1.7559 & -0.5853 \\ -0.5853 & 0.5853 \end{bmatrix}, \quad \mathbf{K} = [-0.8543, -2.5628] \quad \text{and} \quad \mathbf{\Xi} = \begin{bmatrix} 0.7298 & 2.1893 \\ 2.1893 & 6.5678 \end{bmatrix}.$$

The parameters for inner-loop attitude tracking controller (50) are adjusted as $\alpha_1 = 180$, $\alpha_2 = 42$, $\beta_1 = 0.72$, $\beta_2 = 2\beta_1/(1 + \beta_1)$. Note that the selection of the parameters is according to the previous research results and by a simple line search algorithm.³⁹ The total thrust of each follower quadrotor is bounded by $T_{i,\max} = 25 \text{ N}$.

The initial positions and velocities of each follower are specified as $\mathbf{x}_i(0) = \mathbf{y}_i(0) = [70(\Theta - 0.5), 0]^T$, $\mathbf{z}_i(0) = [0, 0]^T$, $i \in \{1, 2, \dots, 6\}$, the original states of the leader are $\mathbf{x}_7(0) = \mathbf{y}_7(0) = [20(\Theta - 0.5), \Theta - 0.5]^T$, $\mathbf{z}_7(0) = [20, \Theta - 0.5]^T$,

where Θ represents a random value between 0 and 1. The initial attitude angle and angular velocity of each vehicle are specified as 0. The real FTCESO estimation error of the outer-loop states and uncertainties for i th aircraft is defined as $\eta_i = [\eta_{i,x}^p, \eta_{i,x}^v, \eta_{i,x}^{\ddot{}}^{\zeta}, \eta_{i,y}^p, \eta_{i,y}^v, \eta_{i,y}^{\ddot{}}^{\zeta}, \eta_{i,z}^p, \eta_{i,z}^v, \eta_{i,z}^{\ddot{}}^{\zeta}]^T$, $i \in \{1, 2, \dots, 7\}$, and the total real estimation error of the whole outer-loop subsystem is $\eta = [\eta_1^T, \eta_2^T, \dots, \eta_7^T]^T$. The trajectory tracking error of each quadrotor is defined as $\varphi_i = [(x_i - h_{i,x} - x_7)^T, (y_i - h_{i,y} - y_7)^T, (z_i - h_{i,z} - z_7)^T]^T$, $i \in \{1, 2, \dots, 6\}$ and the total formation tracking error of the whole multi-quadrotor system is $\varphi = [\varphi_1^T, \varphi_2^T, \dots, \varphi_6^T]^T$. Similarly, the real FTCESO estimation error of the inner-loop subsystem for the i th aircraft is denoted as $\eta_{i,\vartheta} = [\eta_{i,\phi}^{\Theta}, \eta_{i,\phi}^{\Omega}, \eta_{i,\phi}^{\zeta}, \eta_{i,\theta}^{\Theta}, \eta_{i,\theta}^{\Omega}, \eta_{i,\theta}^{\zeta}, \eta_{i,\psi}^{\Theta}, \eta_{i,\psi}^{\Omega}, \eta_{i,\psi}^{\zeta}]^T$, and the total real estimation error of the attitude subsystem for all six followers is $\eta_{\vartheta} = [\eta_{1,\vartheta}^T, \eta_{2,\vartheta}^T, \dots, \eta_{6,\vartheta}^T]^T$. The total attitude tracking error is defined as $\vartheta = [\phi_1 - \phi_1^d, \theta_1 - \theta_1^d, \psi_1 - \psi_1^d, \dots, \phi_6 - \phi_6^d, \theta_6 - \theta_6^d, \psi_6 - \psi_6^d]^T$.

The macroscopic trajectory of the multi-quadrotor system is shown in Fig. 3 in 3-D space, where the leader is marked by a black diamond and the six followers are marked by a blue circle, red square, yellow diamond, cyan hexagram, emerald green pentagram and purple triangle, respectively. It can be seen that the status of the followers and the leader at $t = 10$ s, 15 s, 20 s, 25 s are given and the path of each quadrotor from the initial moment is also depicted. Obviously, the initial positions of the quadrotors are scattered at first and the expected formation is not quite achieved at $t = 10$ s. Then, from $t = 15$ s to 25 s, all six followers keep a stable time-varying parallel hexagram formation with the leader in the center all the time. Fig. 4 shows the total FTCESO real estimation error of the outer-loop subsystem for all seven quadrotors and it is clear that the FTCESOs converge extremely fast and the estimation error can be very small.

We take the outer-loop states of the quadrotor No.1 in the x -axis direction as an example to show the performance of the FTCESO. From Fig. 5, one can see that the states and the uncertainties are perfectly estimated by the FTCESO. Fig. 6 depicts the curve of total formation tracking error of the multi-quadrotor system and it shows that the formation tracking error starts converging extremely small since around $t = 10$ s. Fig. 7 illustrates the convergence of the coupling

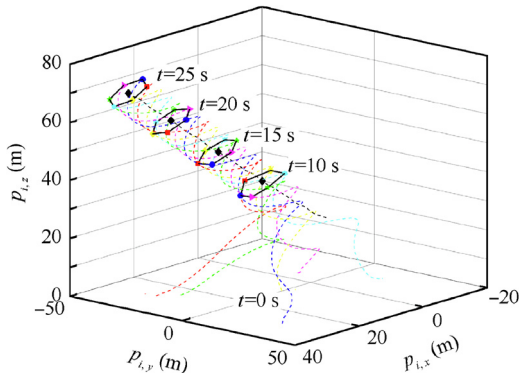


Fig. 3 Trajectory of the multi-quadrotor system.

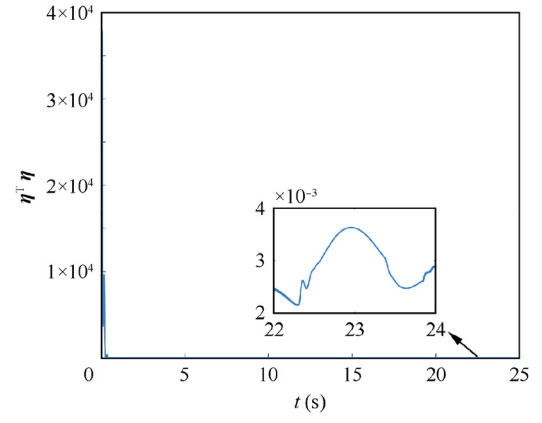


Fig. 4 Total real FTCESO estimation error of outer-loop subsystem for all seven quadrotors.

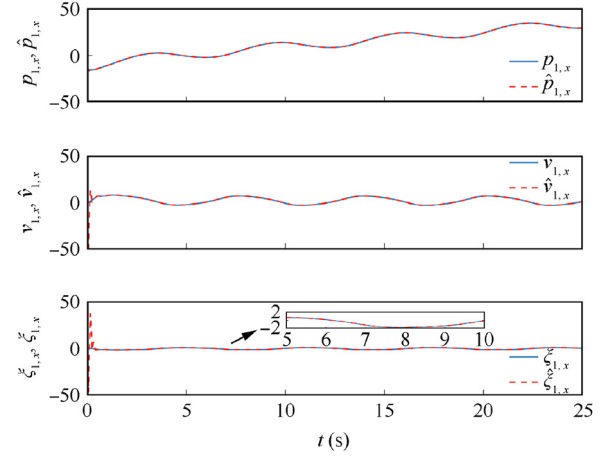


Fig. 5 FTCESO performance of outer-loop subsystem in the x -axis for quadrotor No.1.

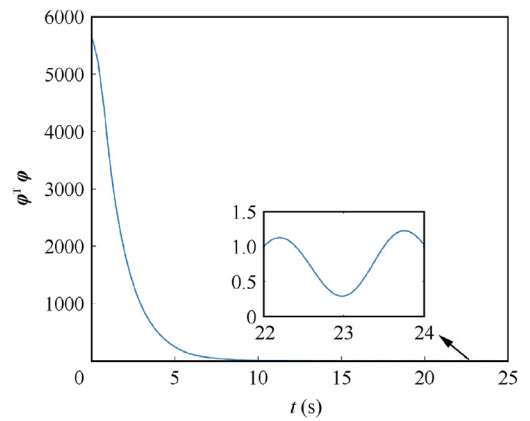


Fig. 6 Total formation tracking error of whole multi-quadrotor system.

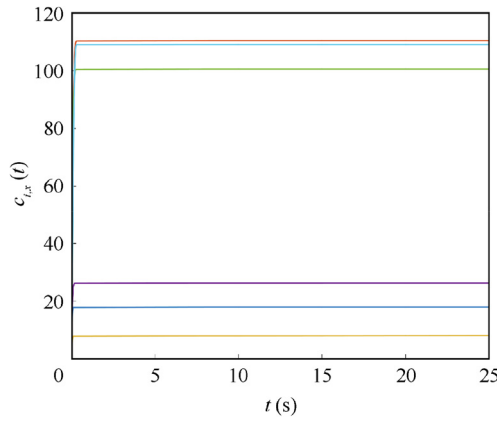


Fig. 7 Coupling weights of all six followers in the x -axis

weights of all the followers in the x -axis direction, that is, Remark 2 is verified.

Fig. 8 displays the total FTCESO real estimation error of the attitude subsystem for six followers and it shows that the FTCESO also works well in the inner-loop subsystem of the multi-quadrotor network. The total attitude tracking error is depicted in Fig. 9 which presents that the inner-loop subsystem converges much faster than the outer-loop subsystem and also the convergence is much smaller.

According to the simulation results above, it is verified that the fully distributed anti-disturbance formation tracking problem is totally solved for the uncertain multi-quadrotor system (10)–(14) under the FTCESO-based fully distributed formation protocol (26) and the FTCESO-based finite-time attitude controller (50). In the following, the improvements of the proposed control scheme are further demonstrated.

4.2. Comparison with the previous work

In this section, the same formation tracking problems in 4.1 are also conducted by an ESO-based anti-disturbance formation tracking protocol proposed in Ref.³³. The improvements and innovations of the control scheme proposed in this paper will be thoroughly demonstrated by the comparisons of simulation below.

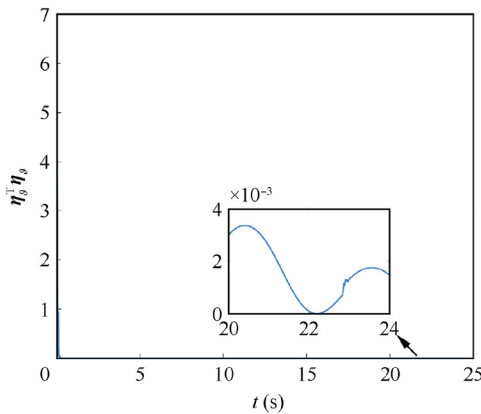


Fig. 8 Total real FTCESO estimation error of inner-loop subsystem for six follower quadrotors

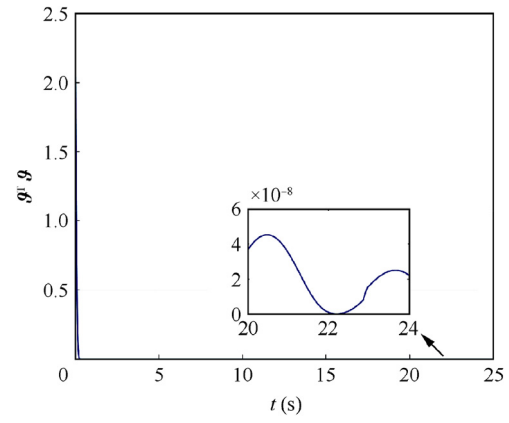


Fig. 9 Total attitude tracking error of whole multi-quadrotor system.

The outer-loop ESOs and ESO-based formation tracking protocol for comparison can be obtained in a second-order form of Eq. (11) in Ref.³³ which can be specified as

$$\begin{cases} \hat{\zeta}_{ij}^p = \hat{\zeta}_{ij}^v + (\hat{h}_{ij}^v - \hat{h}_{ij}^p) + \omega_1 s_{ij} \\ \hat{\zeta}_{ij}^v = \hat{\zeta}_{ij}^{\xi} + \omega_2 s_{ij} + u_{ij} \\ \hat{\zeta}_{ij}^{\xi} = \omega_3 s_{ij} \\ u_{ij} = \mathbf{K}' \sum_{k=1}^6 w_{ik} (\hat{\zeta}_{ij} - \hat{\zeta}_{k,j}) + \mathbf{K}' w_{i7} \hat{\zeta}_{ij} - \hat{\zeta}_{ij}^{\xi} \end{cases} \quad (69)$$

where $i \in \{1, 2, \dots, 6\}$, $j \in \{x, y, z\}$, $\hat{\zeta}_{ij}^p$ is the estimation of $\zeta_{ij}^p = p_{ij} - h_{ij}^p - p_{7,j}$, $\hat{\zeta}_{ij}^v$ is the estimation of $\zeta_{ij}^v = v_{ij} - h_{ij}^v - v_{7,j}$, $\hat{\zeta}_{ij}^{\xi}$ is the estimation of $\zeta_{ij}^{\xi} = \xi_{ij} - \xi_{7,j} - \hat{h}_{ij}^v$. s_{ij} is denoted as $s_{ij} = \sum_{k=1}^6 w_{ik} (\hat{\zeta}_{ij}^p - \hat{\zeta}_{k,j}^p) + w_{i7} \hat{\zeta}_{ij}^p$, and $\hat{\zeta}_{ij}^p$ is defined as $\hat{\zeta}_{ij}^p = \zeta_{ij}^p - \hat{\zeta}_{ij}^{\xi}$. Select $\kappa_0 = 7$, $\tau = 0.04$, $\kappa_{\eta} = 56$. Solving LMIs (14) and (40) in Ref.³³, the corresponding parameters can be obtained as $\omega_1 = 196.3645$, $\omega_2 = 155.8378$, $\omega_3 = 14.4816$ and $\mathbf{K}' = [-5.4286, -7.9465]$.

In Ref.³³, the author only considered the formation tracking control with ESOs for outer-loop subsystem. Thus, for comparison, the comparative attitude tracking controller is based on Eq. (50) only with the estimations produced by the ESO (4) in Ref.³³. Specifically, the comparative ESOs for attitude subsystem (44) can be written as

$$\begin{cases} \hat{j}_{i,1} = \hat{j}_{i,2} + \beta_{01} (j_{i,1} - \hat{j}_{i,1}) \\ \hat{j}_{i,2} = \hat{\zeta}_{ij} + \beta_{02} (j_{i,1} - \hat{j}_{i,1}) + \tau_{ij}, \quad i \in \{1, 2, \dots, 6\}, j \in \{\phi, \theta, \psi\} \\ \hat{\zeta}_{ij} = \beta_{03} (j_{i,1} - \hat{j}_{i,1}) \end{cases} \quad (70)$$

where the observer gains are selected as $\beta_{01} = -150.3628$, $\beta_{02} = -126.8572$, $\beta_{03} = -85.7925$, the parameters $\alpha_1, \alpha_2, \beta_1, \beta_2$ are adopted the same as in Section 4.1. The initial estimation values of ESOs and FTCESOs are the same which can be specified as four fifths of the initial states of the multi-quadrotor system. The comparisons of the formation tracking performance are as below.

Fig. 10 shows that the total real estimation error of FTCESOs for outer-loop subsystem of multi-quadrotor network has

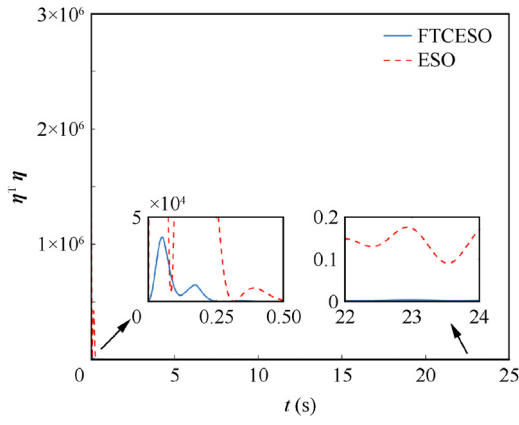


Fig. 10 Comparison of total real estimation error between FTCESO and ESO for outer-loop subsystem.

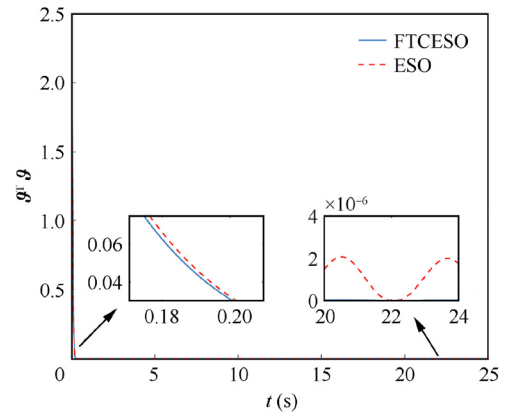


Fig. 13 Comparison of total attitude tracking error between FTCESO-based and ESO-based attitude controller.

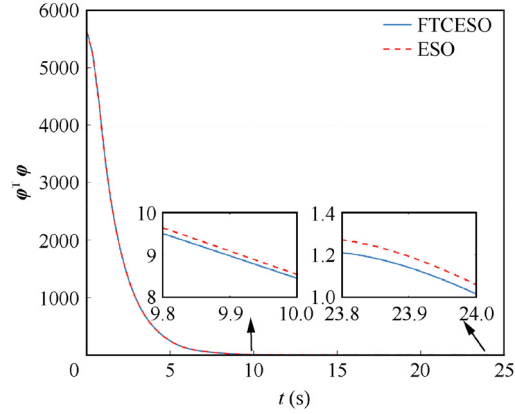


Fig. 11 Comparison of total formation tracking error between FTCESO-based and ESO-based formation protocol.

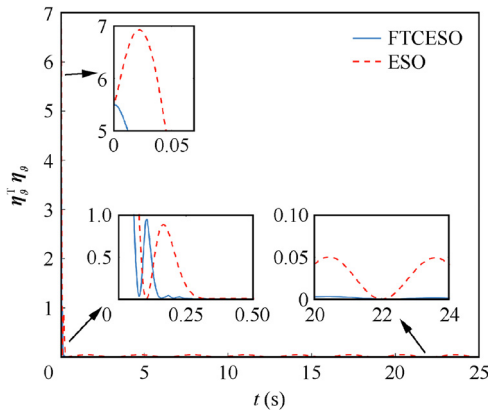


Fig. 12 Comparison of total real estimation error between FTCESO and ESO for inner-loop subsystem.

a much smaller overshoot and convergence than that of ESOs. Besides, the FTCESOs also converge faster than ESOs due to the finite-time convergence characteristic. Fig. 11 further demonstrates that the FTCESO-based formation protocol (26) is of better formation tracking performance than the protocol (11) in Ref.³³.

For the attitude subsystem, the FTCESO still has a better performance in all aspects of overshoot, convergence speed and convergence value, which can be verified in Fig. 12. Because of the better performance of FTCESO, the attitude tracking performance under the FTCESO-based controller is undoubtedly better than that under the ESO-based one. This can also be drawn from Fig. 13.

5. Conclusions

- (1) A novel FTCESO-based fully distributed formation control framework is developed for multi-quadrotor systems with time-varying uncertainties and a directed interaction graph. Under the designed control framework, the time-varying anti-disturbance formation tracking problems can be perfectly solved based only on the aircraft's neighboring output information.
- (2) Based on a detailed algorithm to specify the cooperative control protocol, the feasibility condition to achieve the time-varying anti-disturbance formation tracking for multi-quadrotor systems is derived and the rigorous theoretical analysis of the whole closed-loop system is given.
- (3) By conducting a comparison simulation example, it is clearly shown that, compared with an recently proposed ESO-based formation protocol, this novel FTCESO-based control strategy improves the disturbance rejection and formation tracking performances.
- (4) It is of interest to further study the finite-time formation issues under the proposed control scheme or to extend the results to a switching directed topology case.

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